

Nature of surface electrical conductivity in natural sands, sandstones, and clays

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Abstract. The electrical conductivity of rocks results from conduction through the bulk solution occupying the pores and from surface conduction occurring at the fluid/grain interface. The nature of the surface electrical conductivity of shaly and clean sands and sandstones is examined. Surface conduction is characterized by the specific surface conductance which is the sum of three contributions: (i) Conduction within the electrical diffuse layer, which makes a negligible contribution to the total specific surface conductance. (ii) Conduction in the Stern layer, which is shown to vary significantly with the salinity of the pore fluid at low salinities (10^{-6} to 10^{-3} mol l^{-1}), but becomes independent of salinity at higher salinities. (iii) A mechanism operating directly on the mineral surface, independent of salinity, and perhaps associated with proton transfer. At salinities higher than 10^{-3} mol l^{-1} and at $25^{\circ}C$, the specific surface conductance of quartz and clays is equal to 8.9×10^{-9} S and 2.5×10^{-9} S respectively. Equations describing the influence of surface conductivity and microstructure upon the macroscopic electrical conductivity of sands, sandstones, and shales are also developed.

Introduction

Surface electrical conductivity in rocks is an important parameter commonly used in characterizing hydrocarbon reservoirs [Waxman and Smits, 1968]. It is also closely related to electrokinetic parameters [Pride, 1994] that are beginning to be used to monitor earthquake and volcanic processes. The macroscopic conductivity of a saturated porous medium with insulating grains results from two mechanisms: (i) bulk conduction, which corresponds to electromigration of the ions in the interconnected pore space, and (ii) surface conduction, which corresponds to electrical conduction in the vicinity of the fluid/grain interface [e.g., Pride, 1994; Revil and Glover, 1997]. The question concerning the importance of surface electrical conductivity relative to the electrical conductivity of the electrolyte in the pore space was studied by Waxman and Smits [1968], Watillon and De Backer [1970], and Bussian [1983]. The conclusion reached by these authors was that surface conductivity is an important contribution to macroscopic rock conductivity at low salinity or/and high temperature. However, the nature of surface conductivity remains poorly understood [e.g., Zukoski and Saville, 1986].

Our purpose is to establish the basis of a theoretical framework able to predict surface electrical conductivity in porous media formed with silica and aluminosilicates.

The Specific Surface Conductance of Quartz

The surface of grains which compose the solid matrix of sands, sandstones or clays is typically charged when it is in contact with an electrolyte. The counterions required to balance the mineral surface charge form an "electrical diffuse layer" (EDL). A theoretical model of electrical conduction in the EDL was reported by Revil and Glover [1997]. However, surface conduction in the vicinity of the mineral grains can have other contributions (Pride, personal communication). We write the specific surface conductance Σ_s in the vicinity of a mineral surface as the sum of three contributions:

$$\Sigma_s \equiv \int_0^{\infty} (\sigma(\chi) - \sigma_f) d\chi = \Sigma_s^{EDL} + \Sigma_s^{Stern} + \Sigma_s^{Prot}, \quad (1)$$

where Σ_s^{EDL} , Σ_s^{Stern} , and Σ_s^{Prot} are the contributions to the surface conductivity from ionic conduction in the EDL and in the Stern layer, and associated with proton transfer, respectively. The variable χ is the local distance from the mineral surface considered locally flat, $\sigma(\chi)$ is the local conductivity at the distance χ from the surface, and σ_f is conductivity of the bulk electrolyte. At pH > 3, in presence of a 1:1 salt (e.g., NaCl) plus an acid or a base (e.g., HCl or NaOH), Σ_s^{EDL} is given by [Revil and Glover, 1997, their Eq. (55)],

$$\Sigma_s^{EDL} = 2\chi_d e \left\{ \left(B_{(+)} C_f + B_{H^+} 10^{-pH} \right) \left[\exp \left(-\frac{e\phi_d}{2k_b T} \right) - 1 \right] + \left(B_{(-)} C_f + B_{OH^-} 10^{pH - pK_f} \right) \left[\exp \left(\frac{e\phi_d}{2k_b T} \right) - 1 \right] \right\}. \quad (2)$$

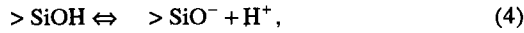
Here C_f and pH are respectively the salinity and the pH of the free electrolyte, e is the elementary charge (1.6×10^{-19} C), k_b is Boltzmann's constant (1.381×10^{-23} J K^{-1}), T is the temperature in Kelvin, $pK_f = -\log_{10} K_f \approx 13.8$ at $25^{\circ}C$ (where K_f is the dissociation constant of water), ϕ_d is the Stern plane potential, i.e., the electrical potential of the inner part of the EDL, and B_i are the equivalent ionic mobilities for ion "i" [Revil and Glover, 1997] $B_i = \beta_i + 2\epsilon_f k_b T / (\eta_f e Z_i)$ where β_i is the ionic mobility of ion "i" which is related to the self-diffusion coefficient D_i by the Nernst-Einstein relationship: $D_i = \beta_i (k_b T) / (e Z_i)$, Z_i is the valence of the ionic species "i", and η_f is the dynamic viscosity of water. The subscripts (+) and (-) refer to the electrolyte cation and anion, respectively. The parameter χ_d is the Debye screening length:

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$$\chi_d = \left(\frac{1}{2} \frac{\epsilon_f k_b T}{10^3 N e^2 (C_f + 10^{-\text{pH}})} \right)^{1/2}, \quad (3)$$

where N is Avogadro's number ($6.02 \times 10^{23} \text{ mol}^{-1}$), ϵ_f is the dielectric constant of water ($\epsilon_f = 80 \epsilon_0$, $\epsilon_0 \cong 8.84 \times 10^{-12} \text{ F}$). The chemical reactions needed to determine the Stern potential in the pH range 5–8 are given by Revil and Glover [1997] as,



where $>$ refers to the mineral lattice, M^+ is the electrolyte cation (e.g., Na^+ or K^+). We note $K_{(-)}$ and K_M the equilibrium constants of Eqs. (4) and (5) [$K_{(-)} = 10^{-7.5}$ at 25°C , *Hiemstra and Van Riemsdijk*, 1990]. In the pH range 5–8, Eqs. (104) and (105) in Revil and Glover [1997] lead to:

$$\varphi_d = \frac{2k_b T}{3e} \ln \Theta, \quad (6)$$

where Θ is defined by,

$$\Theta \equiv \left[8 \times 10^3 N \epsilon_f k_b T (C_f + C) \right]^{1/2} \left(\frac{10^{-\text{pH}} + C_f K_M}{2e \Gamma_s^0 K_{(-)}} \right), \quad (7)$$

where $C \equiv \text{Max}[10^{-\text{pH}}, 10^{\text{pH}-\text{pK}_f}]$, $\text{Max}[a; b]$ is the maximum value of a and b , and Γ_s^0 is the total surface site density. Combining Eqs. (2), (3), and (6), we have,

$$\Sigma_S^{\text{EDL}} = \left(\frac{2 \times 10^3 N \epsilon_f k_b T}{C_f + 10^{-\text{pH}}} \right)^{1/2} \left[(B_{(+)} C_f + B_{\text{H}^+} 10^{-\text{pH}}) (\Theta^{-1/3} - 1) + (B_{(-)} C_f + B_{\text{OH}^-} 10^{\text{pH}-\text{pK}_f}) (\Theta^{1/3} - 1) \right] \quad (8)$$

The Stern layer is formed by ions adsorbed on the mineral surface. These ions can move along the surface when the porous medium is submitted to an electrical field [*O'Konski*, 1960]. The Stern contribution to Σ_S can be written:

$$\Sigma_S^{\text{Stern}} = e \beta_S \Omega_M^0 \Gamma_S^0, \quad (9)$$

where β_S is the equivalent mobility of the adsorbed ions in the Stern layer and Ω_M^0 is the relative surface density of adsorbed ions, given by Eq. (84) of Revil and Glover [1997] as,

$$\Omega_M^0 = \frac{K_M C_f}{10^{-\text{pH}} + K_{(-)} + K_M C_f}. \quad (10)$$

The third contribution, Σ_S^{Prot} , is thought to be directly related to a surface conduction phenomenon such as migration of protons or electrons as suggested by *O'Konski* [1960] (also *S. Pride*, personal communication). This contribution is probably confined in a thin region on or near the surface, and is assumed independent of the salinity but dependent of the total site density of the mineral surface.

We compare the previous equations to the ζ -potential and specific surface conductance data made by *Watillon and de Backer* [1969] on a silica capillary (Fig. 1). Their data results from streaming potential and electrical conductivity measurements ($T \approx 20\text{--}25^\circ\text{C}$, electrolyte: KNO_3). We take for Σ_S^{Prot} the value of the specific surface conductance given by

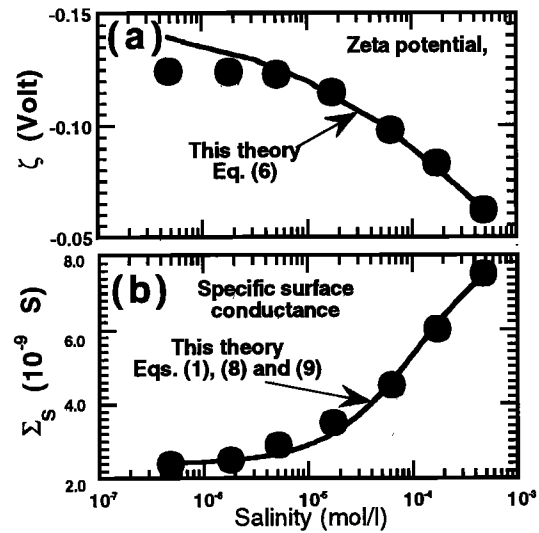


Figure 1. (a) Zeta-potential and (b) specific surface conductance as a function of salinity. The dots represent the experimental data from *Watillon and de Backer* [1970]. The zeta-potential and the specific surface conductance results from streaming potential and electrical conductivity measurements respectively. The solid lines represent the result of the model presented in this paper.

Watillon and de Backer [1969] in pure water: $\Sigma_S^{\text{Prot}} = 2.4 \times 10^{-9} \text{ S}$. We consider $\zeta \approx \varphi_d$ [*Pride*, 1994]. We use the data shown in Fig. 1 to calculate pH, Γ_s^0 , K_M and β_S by doing non-linear fitting of Eq. (1) [in which we have incorporated Eqs. (8) and (9)], and Eqs. (6) to the experimental data. The algorithm used is a standard Newton-Raphson method to solve non-linear equations. We have tested the uniqueness of the solution using a Monte-Carlo procedure in order to assess the sensitivity of the result to the initial guess for the parameters inverted. The initial values for a non-linear inversion are sampled in the possible range of each of the four parameters. The solution is unique, and the theory fits the experimental data very well over the range of salinities for which experimental observations are available ($\sim 10^{-7}\text{--}10^{-3} \text{ mol l}^{-1}$) except for the ζ -potential at the lowest values of salinity. We found from this procedure $\text{pH} = 6.64 \pm 0.04$, $\Gamma_s^0 = 10.0 \pm 0.6 \text{ sites nm}^{-2}$, $K_M = 10^{-(2.77 \pm 0.07)}$ and $\beta_S = (0.40 \pm 0.02) \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$. This result is convincing because the pH value from the inversion compared very well with the measured value reported by *Watillon and de Backer* [1969] in their experiment [6.8 ± 0.5]. The surface site density is close to the values reported by *Hiemstra and Van Riemsdijk* [1990] from geometrical considerations ($\Gamma_s^0 = 9.6 \text{ sites nm}^{-2}$ for the 001 cleavage plane, and $\Gamma_s^0 = 6.0 \text{ sites nm}^{-2}$ for the 101 cleavage plane of quartz crystals). The value of K_M agrees very well with values in the range $10^{-2.5}\text{--}10^{-3.2}$ reported by *Scales et al.* [1990]. The value of β_S is in good agreement with the value of β_S derived in the next section for clay minerals. At salinities such that $C_f \gg \text{Max}(10^{-\text{pH}}/K_M; K_{(-)}/K_M)$ (i.e., larger than $10^{-3} \text{ mol l}^{-1}$), Σ_S reaches a fairly constant value given by $\sim 8.9 \times 10^{-9} \text{ S}$. We found that the EDL contribution to the overall surface conductivity Σ_S^{EDL} is a minor component compared to the other two components (Σ_S^{Stern} and Σ_S^{Prot}), accounting for less than few percent for salinities varying between $10^{-7}\text{--}1 \text{ mol l}^{-1}$.

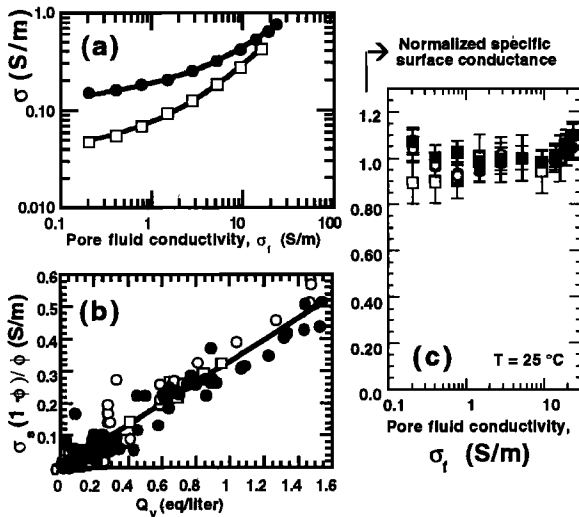


Figure 2. Electrical conductivity of shaly sands. (a) Macroscopic electrical conductivity, σ , as a function of pore fluid electrical conductivity, σ_f [data from Waxman and Smits, 1968]. Eq. (12) is used to determine the electrical formation factor, F , and the surface conductivity, σ_s ($\square$ #WS-17: $F = 46.48 \pm 0.42$, $\sigma_s = (426 \pm 11) \times 10^{-4} \text{ S m}^{-1}$; $\bullet$ #WS-26: $F = 46.95 \pm 0.63$, $\sigma_s = (1521 \pm 29) \times 10^{-4} \text{ S m}^{-1}$). (b) Surface conductivity, σ_s , scaled by $(1-\phi)/\phi$ as a function of the excess charge density per unit pore volume, Q_v , in eq l $^{-1}$ (1 eq = 96320 C). Q_v is estimated from CEC values which are determined by cation exchange experiments. Data from $\square$ Vinegar and Waxman [1984], $\bullet$ Waxman and Smits [1968], $\circ$ Hill and Milburn [1956]. The line represents a linear regression forced through the origin ($r = 0.959$). The slope of this regression gives the surface mobility of the counterions: $\beta_s = 0.51 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$. (c) Normalized specific surface conductance $\Sigma_s(\sigma_f)/\Sigma_s(5.249 \text{ S/m})$ as a function of the pore fluid conductivity, σ_f . The specific surface conductance is calculated using the method described in the text from data for 4 sands samples with high clay content [#17, #23, #24 and #26, Table 7 from Waxman and Smits, 1968]. The clay fractions are: $\square$ WS#17 (K:100%, I: 0%, S: 0%), $\circ$ WS#23 (K:8%, I: 8%, S: 84%), $\bullet$ WS#24 (K:12%, I: 8%, S: 84%), $\blacksquare$ WS#26 (K:0%, I: 0%, S: 100%), (K, kaolinite; I, illite; and S, smectite).

Specific Surface Conductance of Clay Minerals

We have taken a “microscopic approach” in the previous section of this paper for quartz because it is relatively easy to have measurements of electrical conductivity (and ζ -potential) in a capillary with smoothed walls of silica. Such data are available in the literature [e.g., Watillon and De Backer, 1970]. In this case, the microgeometry is known and can be removed from the macroscopic measurements. For clays, the problem is far more complex due to the geometry of the clay particles themselves. We consider here a macroscopic approach in which the microgeometry is explicitly incorporated in the equation for the macroscopic electrical conductivity. The validity of this approach is tested using a large database of experimental results. We combine the work of Bussian [1983] based on the Hanai-Bruggeman equation with the work of Revil and Glover [1997] which takes into account the different behavior of anions and cations. We note

$t_{(+)}^f$ and $t_{(-)}^f = 1 - t_{(+)}^f$, the fraction of electrical current carried in the free electrolyte by the cations and the anions respectively. These parameters are called the “Hittorf numbers” of the cations and anions in the free electrolyte [e.g., Revil and Glover, 1997] and can be calculated by $t_{(+)}^f = \beta_{(+)}^f / (\beta_{(+)}^f + \beta_{(-)}^f)$, where $\beta_{(\pm)}^f$ (in $\text{m}^2 \text{ s}^{-1} \text{ V}^{-1}$) are the mobilities of the ions in the bulk electrolyte [for NaCl, $t_{(+)}^f \approx 0.38$]. Similarly, we note $t_{(\pm)}^s$ the fraction of current carried by anions and cations for surface conduction. The macroscopic electrical conductivity, σ , of a saturated sand or sandstone is given by $\sigma = \sigma_{(+)} + \sigma_{(-)}$ where each ionic contribution $\sigma_{(\pm)}$ is given by the Hanai-Bruggeman equation [Bussian, 1983]:

$$\sigma_{(\pm)} = \frac{t_{(\pm)}^f \sigma_f}{F} \left(1 - \frac{t_{(\pm)}^s \sigma_s}{t_{(\pm)}^f \sigma_f} \right)^m \left(1 - \frac{t_{(\pm)}^s \sigma_s}{\sigma_{(\pm)}} \right)^{-m}, \quad (11)$$

where σ_s denotes surface conductivity, F is the electrical formation factor related to the porosity ϕ by Archie’s law $F = \phi^{-m}$, m is the cementation exponent [e.g., Waxman and Smits, 1968]. For a granular porous medium, σ_s is related to the specific surface conductance by $\sigma_s = (2 \Sigma_s / R)$ where R is the radius of the grains [O’Konski, 1960]. Taking $m \approx 2$ in Eq. (11) [which is a good approximation for the cementation exponent, e.g., Waxman and Smits, 1968], and $t_{(+)}^f = 1$ (and therefore $t_{(-)}^f = 0$), i.e., the counterions are essentially cations [Revil and Glover, 1997], Eq. (11) can be rewritten as a second degree equation in $\sigma_{(\pm)}$ which can be easily solved to give σ :

$$\sigma = \frac{\sigma_f}{F} \left[1 - t_{(+)}^f + F \xi + \frac{1}{2} (t_{(+)}^f - \xi) \left(1 - \frac{\xi}{t_{(+)}^f} + \sqrt{\left(1 - \frac{\xi}{t_{(+)}^f} \right)^2 + \frac{4 F \xi}{t_{(+)}^f}} \right) \right] \quad (12)$$

where the dimensionless parameter ξ is defined by [Kan and Sen, 1987],

$$\xi \equiv \frac{\sigma_s}{\sigma_f} \approx \frac{2}{3} \left(\frac{\phi}{1-\phi} \right) \frac{\beta_s Q_v}{\sigma_f}. \quad (13)$$

Here β_s is the surface mobility of the counterions, and Q_v is the excess surface charge per unit pore volume, which is related to the “Cation Exchange Capacity” (CEC) [Waxman and Smits, 1968] by $Q_v \equiv \rho_m [(1-\phi)/\phi] \text{CEC}$, where ρ_m is the grain density (in kg m^{-3}). The CEC (in C kg^{-1} of matrix weight) indicates the maximum number of surface exchangeable cations per unit mass of shaly rock. It strongly depends on clay mineral type. Patchett [1975] gives: $\text{CEC}(\text{kaolinite}) = 0.04 \pm 0.02 \text{ meq g}^{-1}$, $\text{CEC}(\text{illite}) = 0.22 \pm 0.02 \text{ meq g}^{-1}$ and $\text{CEC}(\text{smectite}) = 1.5 \pm 0.5 \text{ meq g}^{-1}$ ($1 \text{ meq g}^{-1} = (e N) \text{ C kg}^{-1} = 96320 \text{ C kg}^{-1}$). For a mixture of clay, we can take an arithmetic average of the CEC weighted by the corresponding mass fraction of each clay mineral. Eq. (12) is a new equation relating the macroscopic electrical conductivity to the microstructural parameters, F and ϕ , and surface properties, $\beta_s Q_v$. Eq. (12) is compared with experimental data in Fig. 2a. Clearly the theoretical predictions accurately match the experimental data at all electrolyte salinities. It should be noted that this is achieved with a single constant value for the formation factor and the surface conductivity.

We test Eq. (12) against experimental data at 25°C from databases in Hill and Milburn [1956], Waxman and Smits [1968], and Vinegar and Waxman [1984]. We use all the 129 samples from these databases with $0 \leq Q_v \leq 1.60 \text{ eq l}^{-1}$ ($1 \text{ eq l}^{-1} = 96.32 \times 10^6 \text{ C m}^{-3}$). We calculate F and σ_s by fitting Eq. (12)

non-linearly to the experimental rock conductivity as a function of fluid conductivity data for each sample. Surface conductivity is commonly plotted directly as a function of Q_V [e.g., Waxman and Smits, 1968; Bussian, 1983]. If this is done for the surface conductivity derived above, the data show a considerable degree of scatter. By contrast, our model [Eqs. (12) and (13)] implies a plot of $\sigma_S(1-\phi)/\phi$ as a function of Q_V as shown in Fig. 2b. We use linear regression of the data and Eq. (13) to derive $\beta_S(25^\circ\text{C}) = 5.14 \times 10^{-9} \text{ m}^2\text{s}^{-1}\text{V}^{-1}$. The ionic mobility of Na^+ at 25°C in water is $\beta_{(+)}'(25^\circ\text{C}) = 5.2 \times 10^{-8} \text{ m}^2\text{s}^{-1}\text{V}^{-1}$ [Waxman and Smits, 1968]. The ratio of the surface mobility to that in the bulk electrolyte for Na^+ is therefore 0.098. The relationship between the specific surface conductance Σ_S and the surface mobility β_S is $\Sigma_S = e\beta_S\Gamma_S^0$ [Patchett, 1975]. Using $\Gamma_S^0 = 3$ charges per nm^2 [Patchett, 1975], the specific surface conductance of clays is $\Sigma_S(25^\circ\text{C}) = 2.47 \times 10^{-9} \text{ S}$.

We examine the dependence of the surface conduction on salinity. The database of Waxman and Smits [1968, Table 7] is well suited for that purpose since it covers a large range of Q_V and of pore fluid conductivities (~ 0.2 to 20 S m^{-1}). First, we calculate F and σ_S by fitting Eq. (12) non-linearly to the experimental conductivity data as a function of fluid conductivity in the range 2.8 – 16 S m^{-1} for each sample. Second, keeping the formation factor constant, we recalculate the surface conductivity σ_S for the complete salinity range using Eq. (12). Consequently, we can reconstruct the dependence of the surface conductivity on the pore fluid conductivity in the range 0.2 to 20 S m^{-1} (Fig. 2c). This figure shows that the surface conductivity is approximately constant within the salinity range investigated (~ 0.02 – 3 mol l^{-1}).

The independence of the specific surface conductance with salinity for salinities $>10^{-2} \text{ mol l}^{-1}$ has physical implications for the conduction mechanism. Indeed, independence on salinity is only possible if the conduction takes place behind the Stern plane as suggested by Zukovski and Saville [1986]. In the dynamic Stern layer model of Zukovski and Saville [1986], the ratio between the mobility in the Stern layer and the mobility in the free electrolyte for K^+ is 0.10 independent of salinity. Our results provide the same ratio for the mobility of Na^+ , and demonstrate the independence of surface mobility with the salinity (Fig. 4). These results strongly suggest a mechanism for surface electrical conduction in clay-rich sediments similar to that postulated by Zukovski and Saville [1986]. Application of Eq. (8) to clay minerals shows that the contribution of the EDL to Σ_S can be safely ignored. The consistency of the results found with the macroscopic approach that we used and the striking ability of Eq. (12) to describe electrical conductivity data (Fig. 2) makes us confident in the ability of Eq. (12) to account correctly for the microstructure in the electrical conductivity problem applied to shaly sands.

Conclusion

The conclusions of this paper are the following. (1) The contribution of the electrical diffuse layer to the surface conductivity can be neglected for silica and aluminosilicates.

(2) the main contribution to surface conductivity is probably due to counterions located in the Stern layer with surface mobilities approximately ten times smaller than in the bulk electrolyte. (3) A third contribution is also required for quartz, which is salinity independent, and possibly related to proton transfer directly on the mineral surface. (4) A new electrical conductivity equation has been developed which can be used to quantify the influence of surface conductivity at the pore-grains interface and the microstructure upon the macroscopic electrical conductivity of natural sands and sandstones.

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